# Attacks

# Attacks

- Motivations
- Target categories
- Attacking each target category

# Motivations

1. Money
2. Stealing of information
3. Disruption of operations

- Money is by far the **most frequent** motivation

# How to obtain money (I)

- **MANY** (creative) ways
  - Banking credentials stolen and used
  - Credentials stolen and sold
  - Long term cookies stolen and sold
  - …
  - Remote Access Trojans (remotely controllable malware) installed and sold / rented
  - …

- Victim **not** aware of what happened

# How to obtain money (II-a)

- Many (very creative) ways
  - …
  - Steal data and ask ransom for not making it public
  - Encrypt data and ask ransom for decrypting it (**ransomware**)

# How to obtain money (II-b)

- Steal data and ask ransom for not making it public
- Encrypt data and ask ransom for decrypting it (ransomware)
- **Huge** societal problem
  - Attack cost relatively low
  - Potential ROI (Return on Investment) huge

  ⇒ Lot of potential attackers

  - Anonymous payments worldwide
  - Data is crucial to "every organization"
  - Worldwide connectivity

  ⇒ Every organization is a potential target

# Target Categories (I)

1. Organizations
2. Industrial Control Systems (ICS)
3. Single individuals

- Organization = "wherever there are servers and data"
- ICS = "sensors and actuators"

# Organization

1. Organizations
   ("wherever there are servers and data")

- **Any** kind of organization
  - Hospitals
  - Administrative part of manufacturing companies
  - …

# Organization vs ICS

- Administration
- Logistics
- Payroll
- Sales / Purchasing
- Warehouse
- ...
- Email / Web
- ...

"Sensors" and "Actuators"

Organization
("IT part of an industry")

ICS

# Target Categories (II)

- You can make **lot of money** with one Organization / lot of Single individuals
- Making money by attacking an ICS is much more difficult

- Attacks to Organizations / Single individuals ⇒**very frequent**
- Attacks to ICS ⇒ **rare**

# Keep in mind

- Attacks are a **professional** activity
- Huge gains justify **huge investments**

- `search "conti diaries part 2"`
  - Tens of people hierarchically structured
  - Work around the clock
  - Teams update malware every 4 hours (update time of Windows Defender)

# "Conti Tech Start-up"

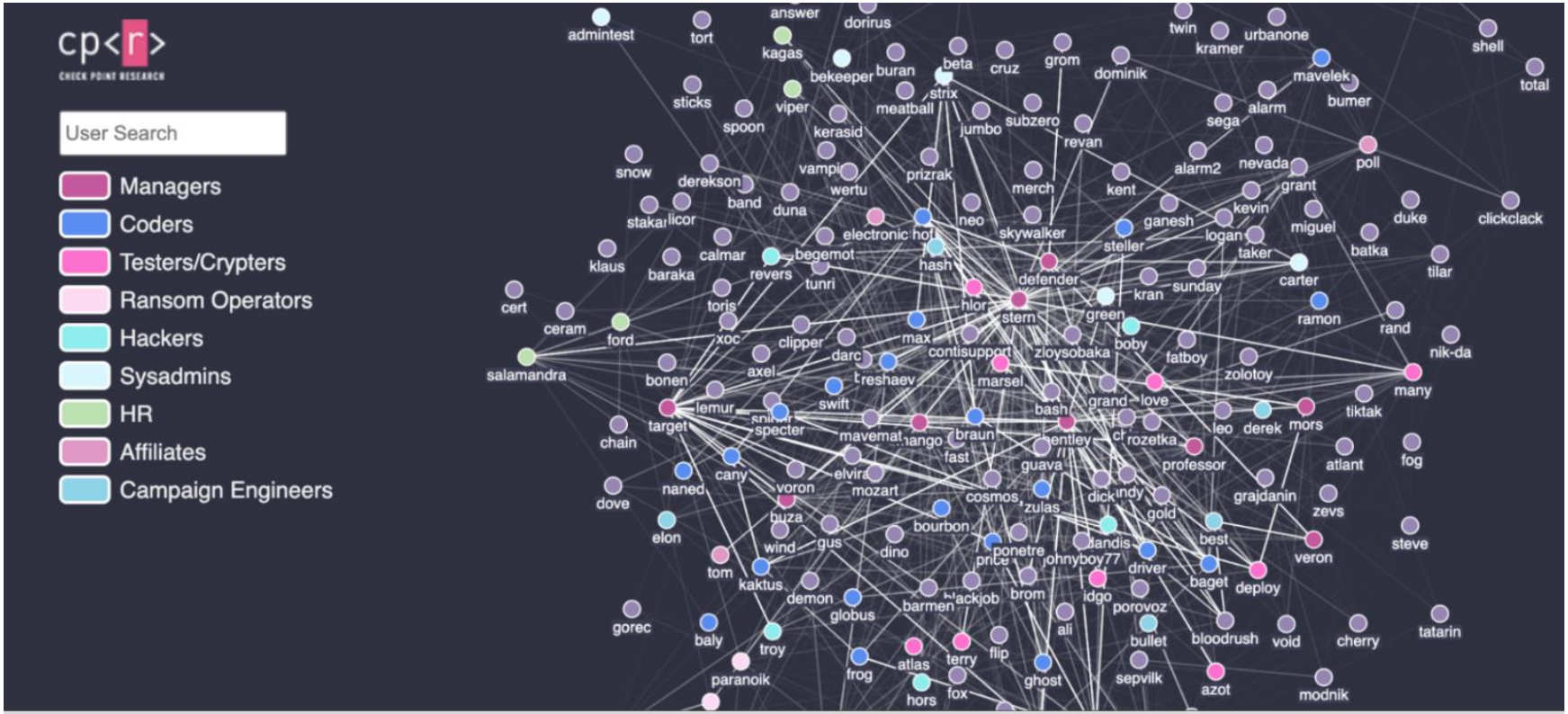



Immagine da research.checkpoint.com

# Our next steps

- Attacks against **organizations**
  - Lateral movement
- …against **single individuals**
- …against **ICS**

# Attacking an Organization

# Attacking an Organization

- Several **phases**, each of several **steps**
- From **minutes** to **months**

- Several **models** for **describing** attack phases
  - Kill chain (first widely used)
  - …
  - MITRE ATT&CK ("the" model today)

# MITRE ATT&CK (I)

- Currently **the** reference framework
- Built upon **observations** of **many real attacks**

- **14** phases (called "**Tactics**")
- Several ways for executing each phase ("**Techniques**")

# MITRE ATT&CK Matrix

# MITRE ATT&CK (II)

- Periodically **updated** to reflect more recent/accurate knowledge
  - October 2022: v12
  - April 2023: v13
  - October 2023: v14

- Three variants
  - Enterprise (may be specialized for Windows, Linux, Cloud,...)
  - Mobile (may be specialized for Android / iOS)
  - ICS
- Reports describe campaigns in terms of MITRE ATT&CK

# Example

## Iranian Government-Sponsored APT Actors Compromise Federal Network, Deploy Crypto Miner, Credential Harvester

CYBERSECURITY & INFRASTRUCTURE SECURITY AGENCY

**Last Revised:** November 25, 2022 **Alert Code:** AA22-320A

### MITRE ATT&CK TACTICS AND TECHNIQUES

See table 1 for all referenced threat actor tactics and techniques in this advisory, as well as corresponding detection and/or mitigation recommendations. For additional mitigations, see the Mitigations section.

# "Gain foothold" (I-a)

- Initial Access

The adversary is **trying to get into your network**.

Techniques that use various entry vectors to gain their **initial foothold** within a network.

# "Gain foothold" (I-b)

- Initial Access

  - **Drive-by Compromise** User visiting a website over the normal course of browsing. Vulnerability exploitation.
  - **Exploit Public-Facing Application** Vulnerability exploitation in an Internet-facing computer or program (e.g., web site)
  - **Phishing**. Malicious attachments or links in emails
  - **Valid Accounts**. Abuse of compromised credentials

  (+5 Techniques) MITRE ATT&CK

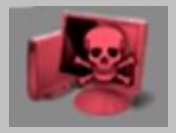
# "Gain foothold" (II)

- Initial Access
- Execution
- Persistence

**Execution** techniques that result in **adversary-controlled** code running within the organization
(12 techniques)

**Persistence** techniques for **keeping access** to systems **across restarts**, **changed credentials**, and other interruptions that could cut off their access.
(19 techniques)

# Scenario so far



# Command & Control (C&C)

- Initial Access
- Execution
- Persistence
- C&C (**Command** & **Control**)
- [illegible]

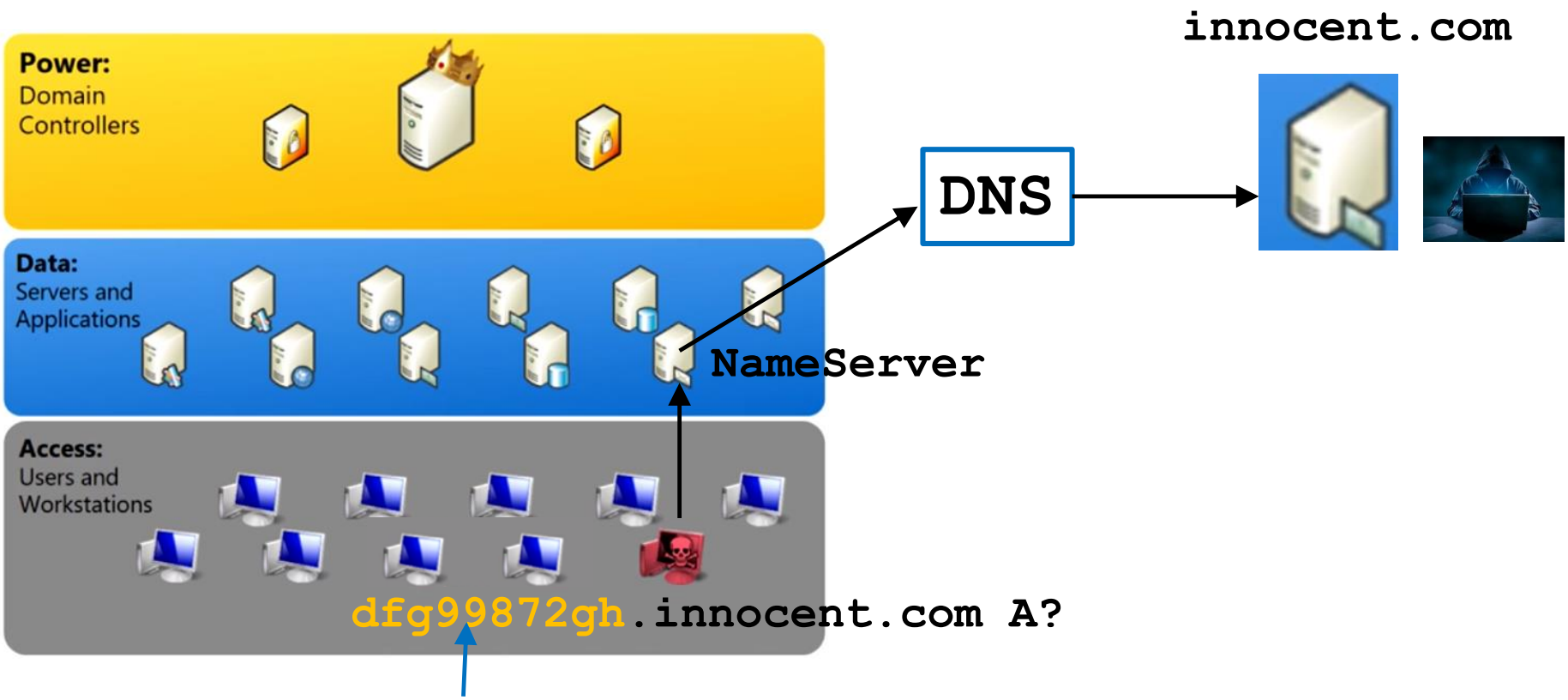
Techniques that adversaries may use to **communicate with systems under their control** within a victim network.

Adversaries commonly attempt to **mimic normal**, expected traffic to **avoid detection**.

**Location** of the adversary must be **obfuscated**.

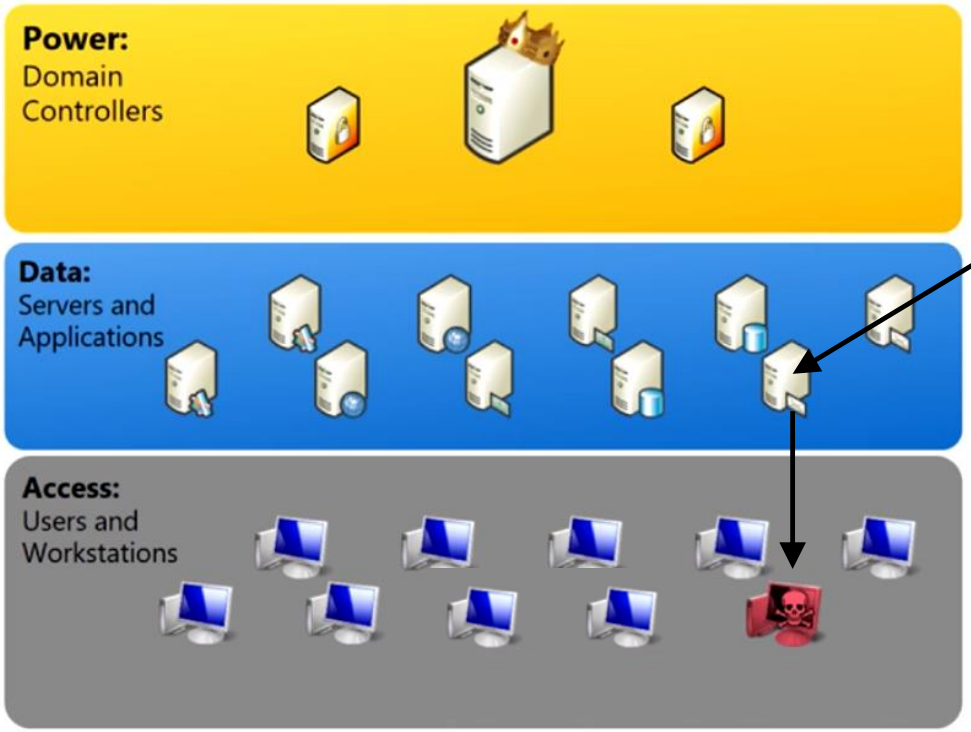
(16 Techniques)

# Example (outline): DNS Tunneling (I)

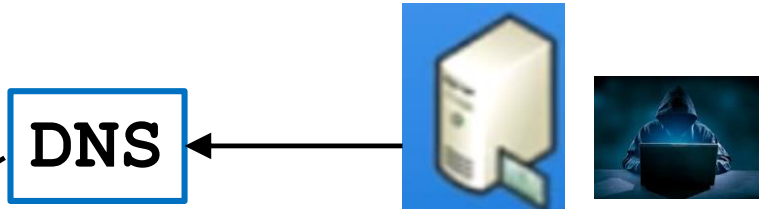



Encodes a **request** message to attacker

# Example (outline): DNS Tunneling (II)




innocent.com



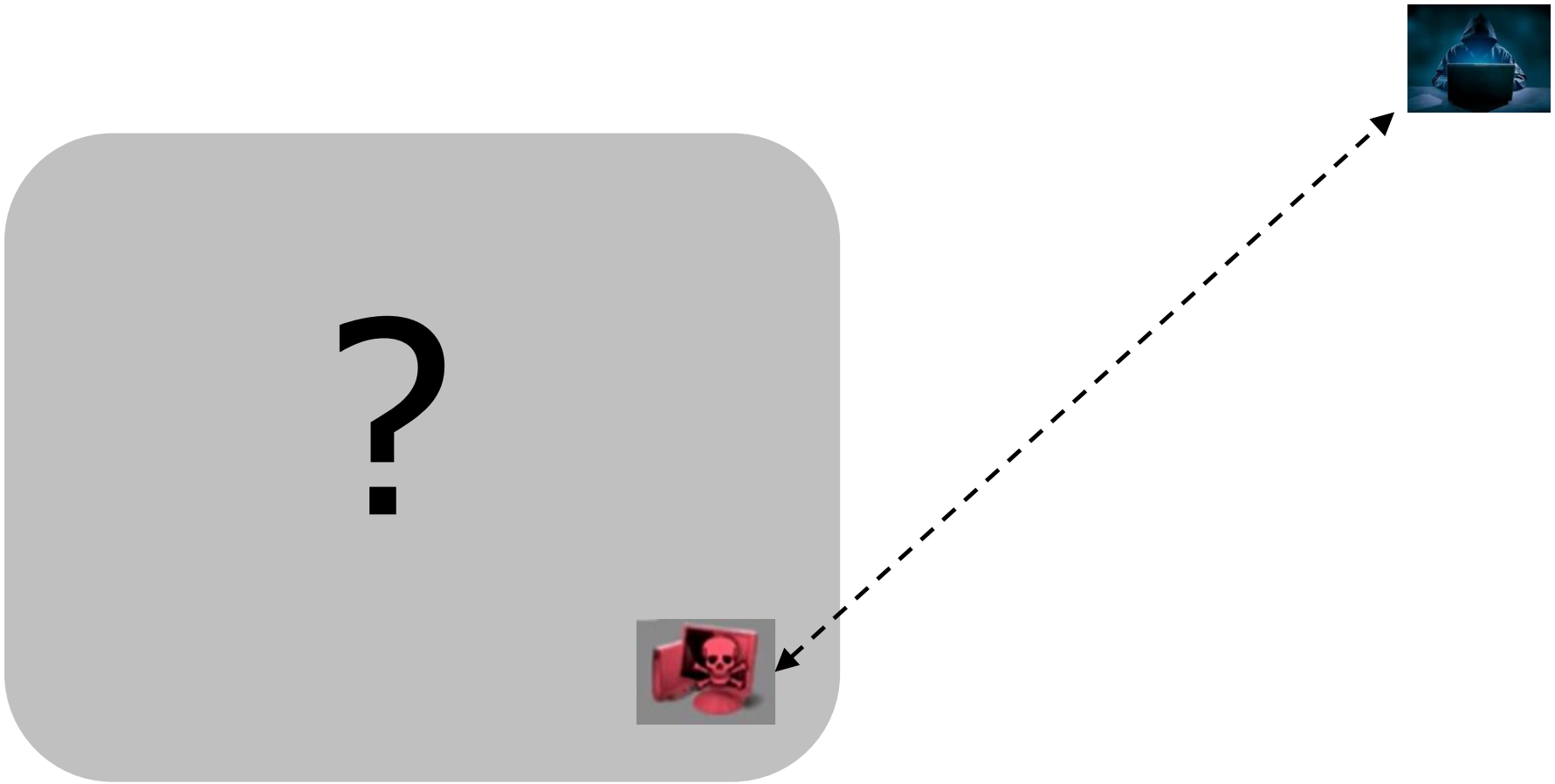

dfg99872gh.innocent.com

CNAME

hhjsd67.innocent.com

Encodes a **response** message from attacker

# Scenario so far



# "Look around"

- Initial Access
- Execution
- Persistence
- C&C (Command & Control)
- Discovery

Techniques to **gain knowledge** about the internal environment and decide how to act

- Networks, Hosts, Devices
- Applications
- Users, Groups, Access Rights
  (29 Techniques)

# **Example:** `nmap`

- Nmap ("Network Mapper") is an open source tool for **network exploration** and **security auditing**.
- It was designed to rapidly scan large networks, although it works fine against single hosts.
- Nmap uses raw IP packets in novel ways to determine
  - what **hosts** are available on the network,
  - what **services** (application name and version) those hosts are offering,
  - what **operating systems** (and OS **versions**) they are running,
  - what type of **packet filters/firewalls** are in use,
  - and dozens of other characteristics.

- Usually quite noisy...

# "Walk around"

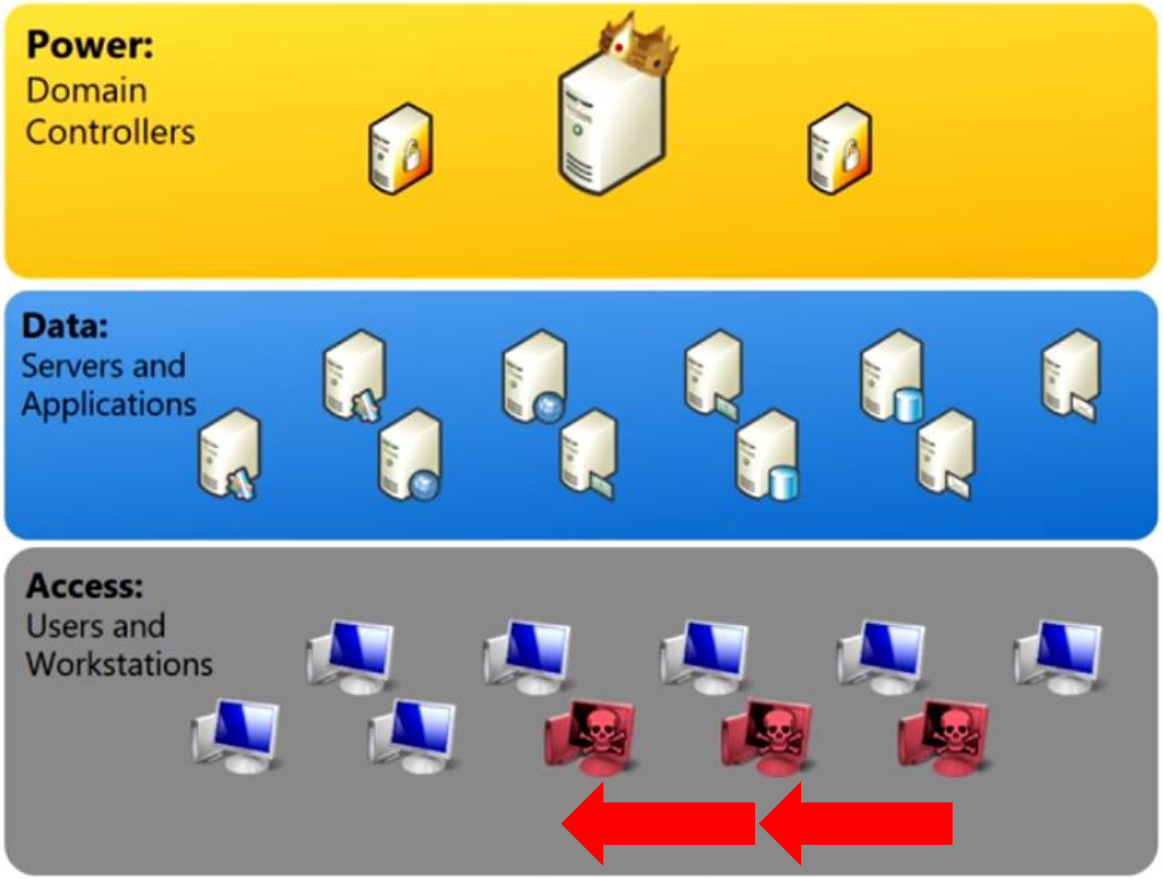
- Initial Access
- Execution
- Persistence
- C&C (Command & Control)
- Discovery
- Lateral movement

Techniques to **enter** and **control** remote systems

(9 Techniques)

We will discuss this phase later

# Lateral Movement



# Privilege Escalation (I)

- Initial Access
- Execution
- Persistence
- C&C (Command & Control)
- Discovery
- Lateral movement
- Privilege escalation

Techniques for **gaining higher-level permissions** on a system or network

(13 Techniques)

# Privilege Escalation (II-a)

**Privilege Escalation**
13 techniques

- **Exploitation for privilege escalation**
  Adversaries may exploit software **vulnerabilities** in an attempt to elevate privileges.

- **Valid Accounts**
  Adversaries may obtain and abuse **credentials of existing accounts**. Adversaries may choose not to use malware or tools in conjunction with the legitimate access those credentials provide to make it harder to detect their presence.

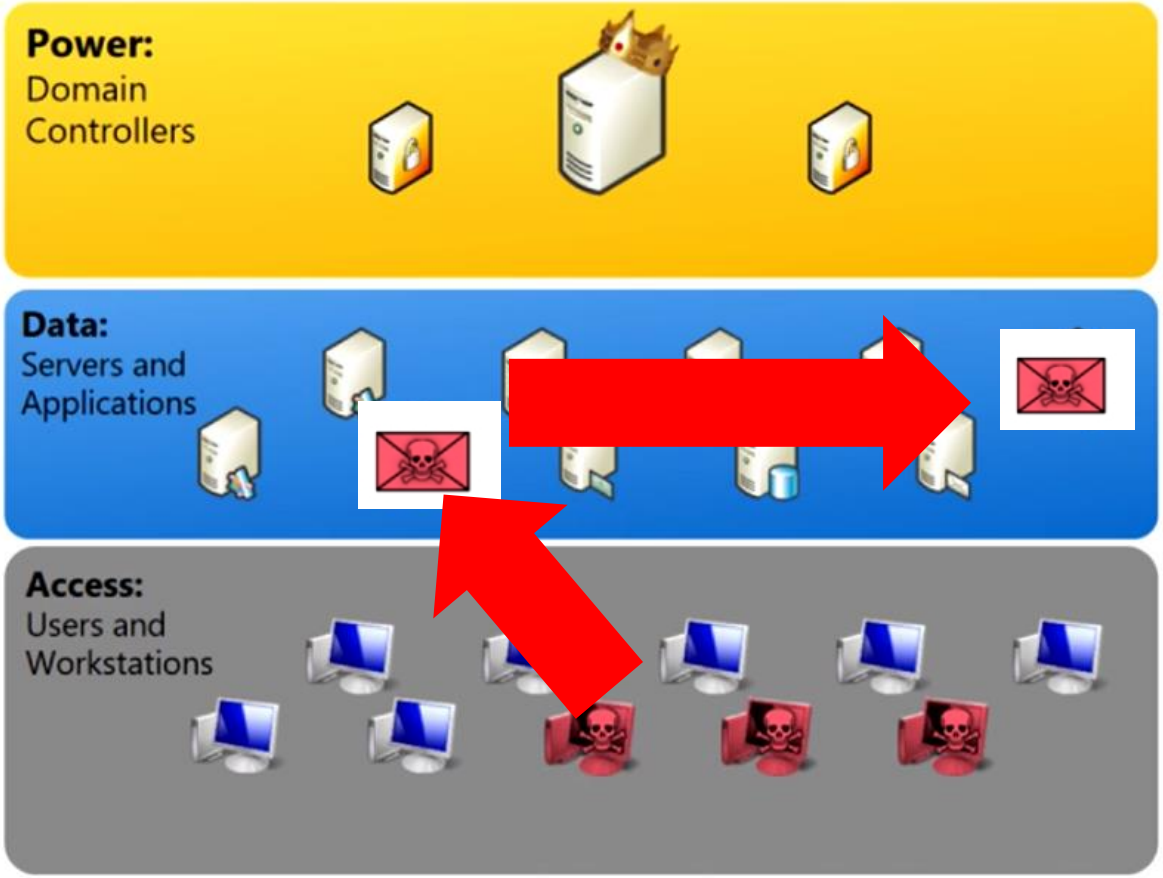
# Privilege Escalation (II-b)

**Privilege Escalation**
13 techniques

- **Domain policy modification**
  Adversaries may **modify the configuration** settings of a domain to escalate privileges in domain environments... Since domain configuration settings control many of the interactions within the Active Directory (AD) environment, there are a great number of potential attacks that can stem from this abuse.
- **...**

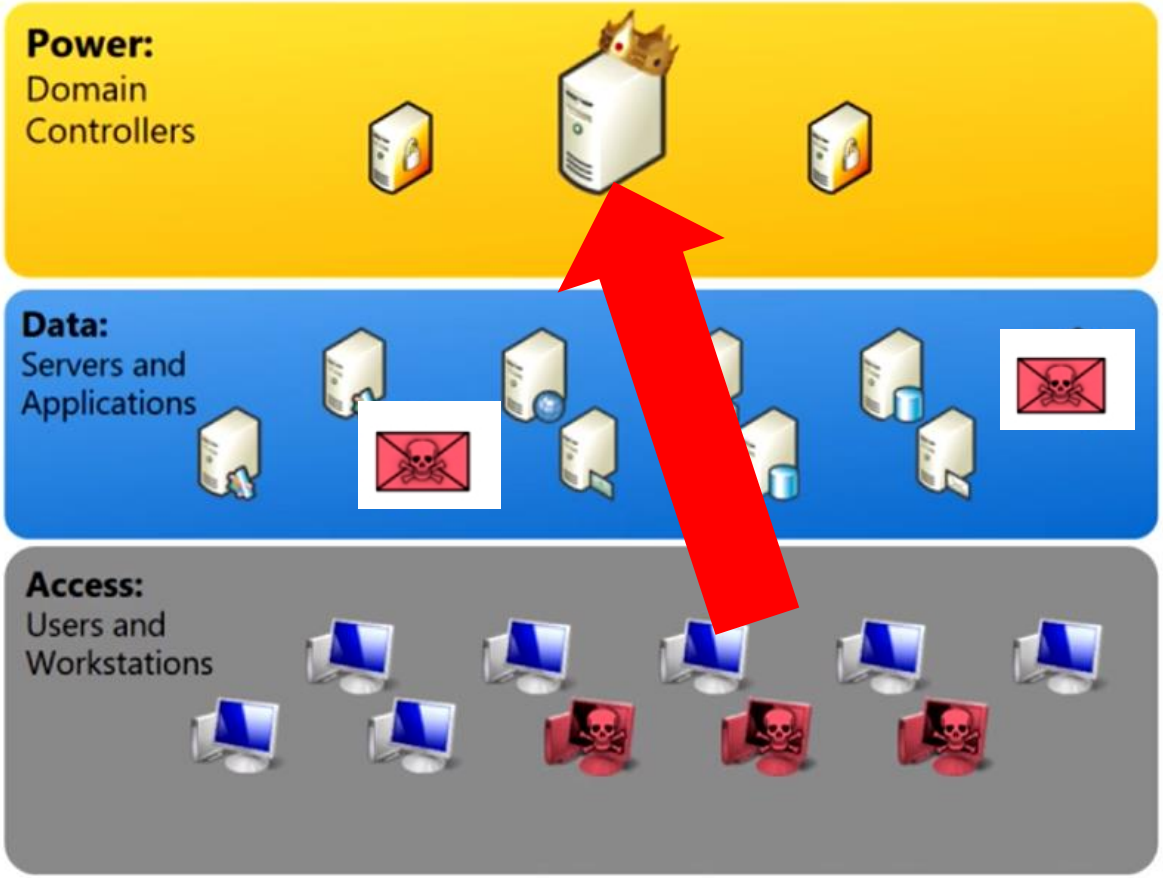
(+10 more techniques)

# Lateral Movement after Privilege Escalation (I)




- Attacker can access "data"
- Which data and which access rights will depend on the available credentials

# Lateral Movement after Privilege Escalation (II)




**Total Catastrophe**

# Exfiltrate

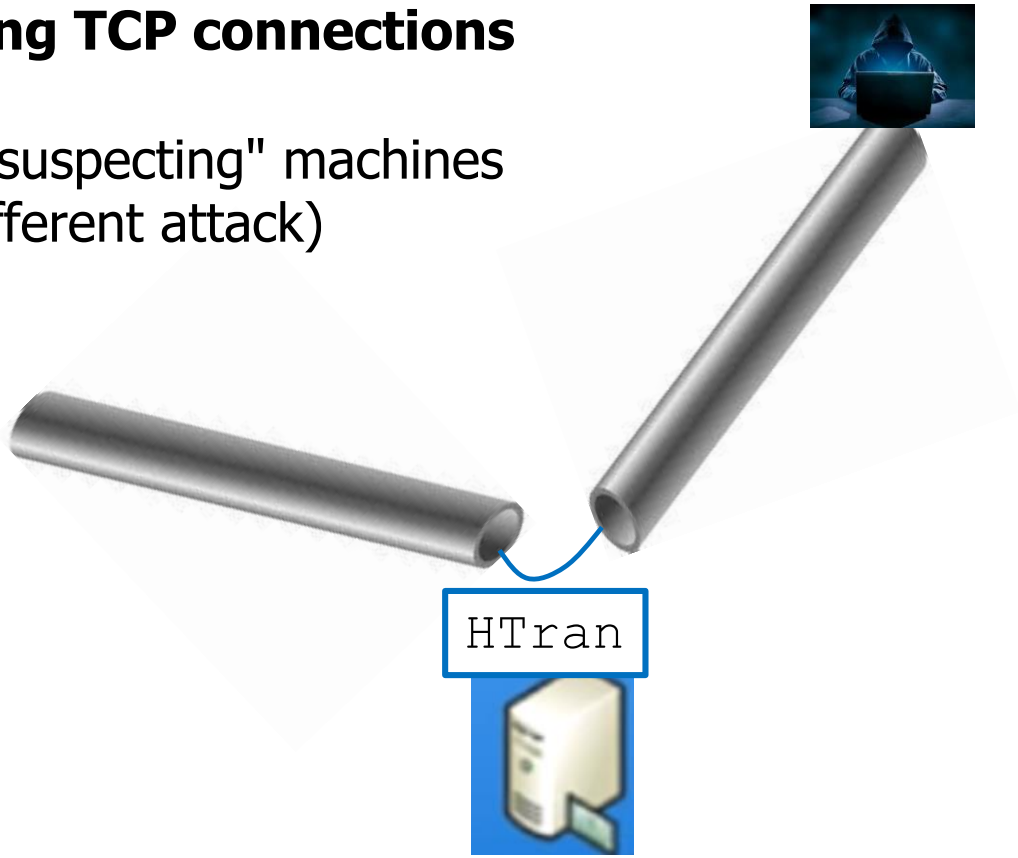
- Initial Access
- Execution
- Persistence
- C&C (Command & Control)
- Discovery
- Lateral movement
- Privilege escalation
- Exfiltration

- **Steal data**
- Transferring it over their C&C channel or an alternate channel
- Compression, Encryption, Size limits

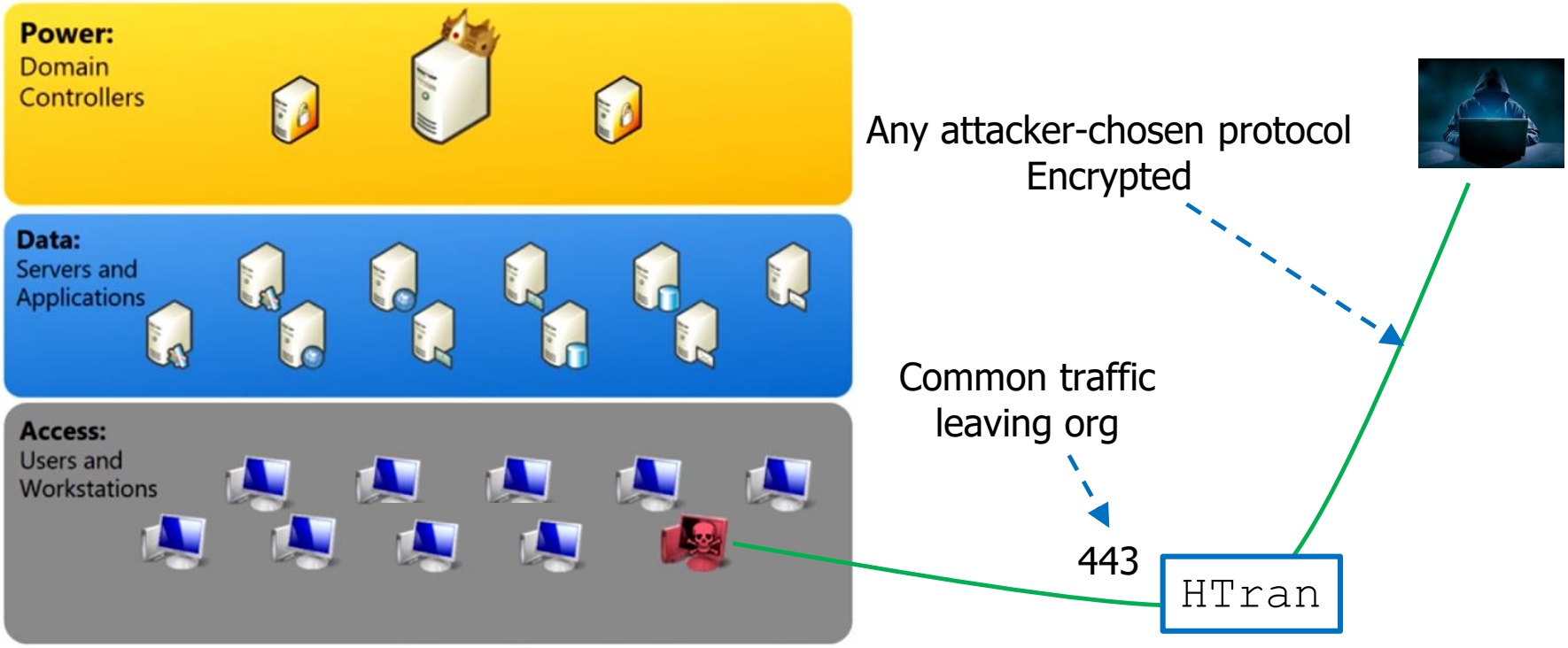
(9 Techniques)

# Example: `HTran` (I)

- Tool for **proxying TCP connections**
- Installed on "unsuspecting" machines (with a prior, different attack)

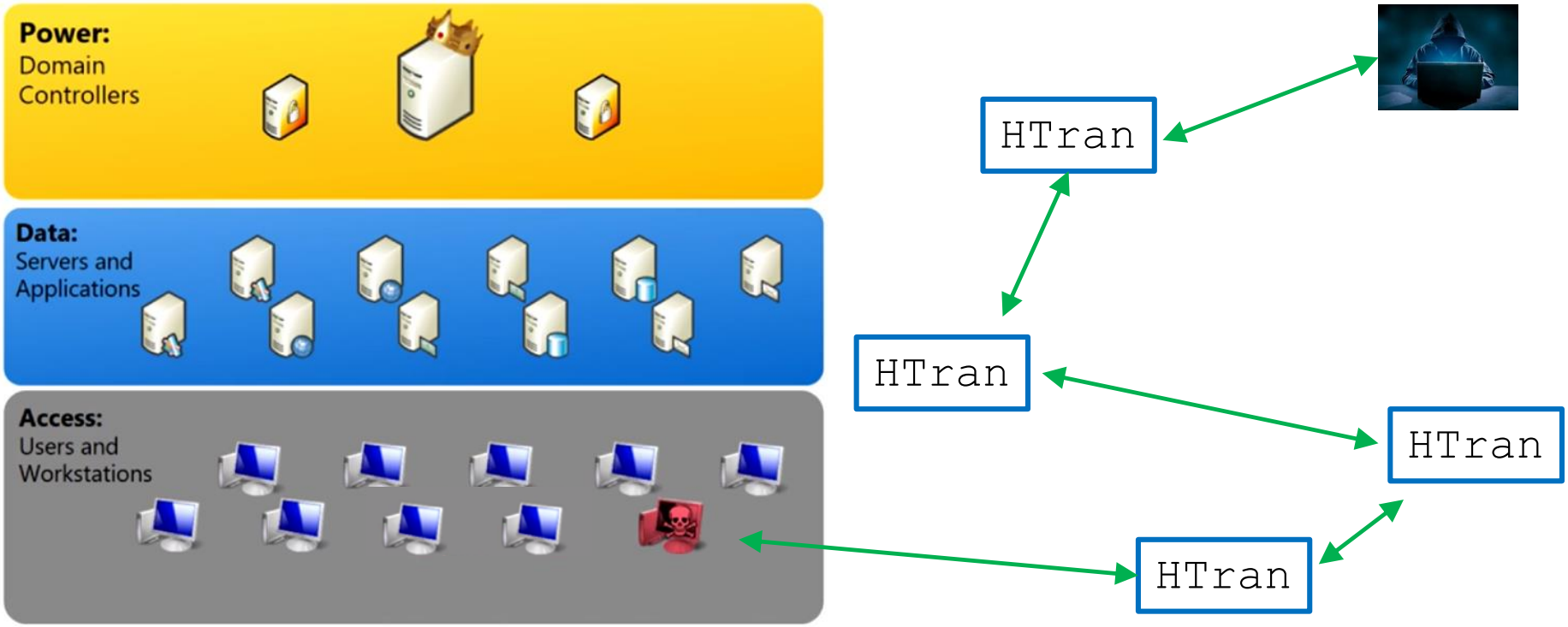


# Example: `HTran` (II-a)




- "By using HTran in this way, the threat actor... **several months** without being detected."

# Example: `HTran` (II-b)




- "By using HTran in this way, the threat actor... **several months** without being detected."

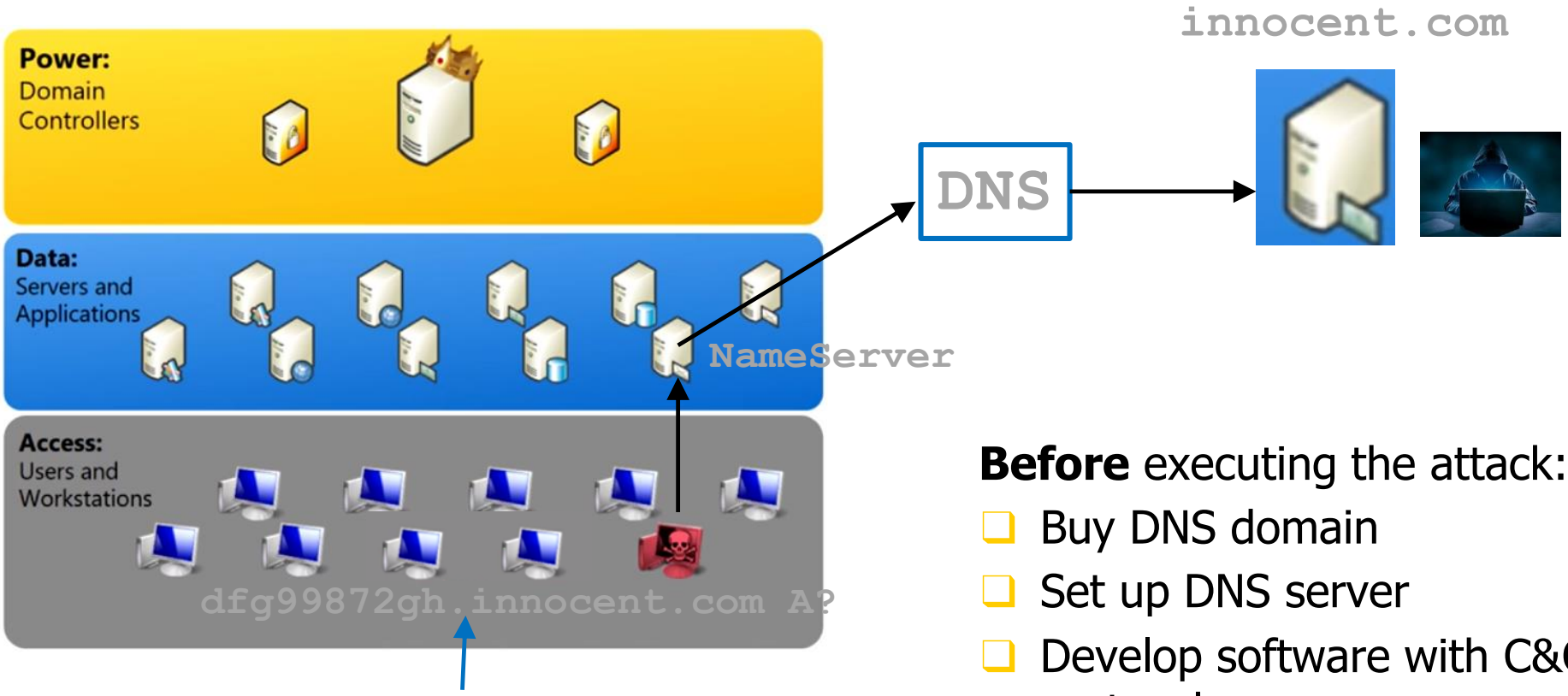
# One sequence does not fit all

- ...
- ~~Exfiltration~~ Impact

- **Impact** Manipulate, interrupt, or destroy your systems and data (≈~~secrecy~~, availability, integrity)
- Examples: **ransomware**, web defacement, disk wiping, **...**

(13 techniques)

# REMIND




Encodes a **request** message to attacker

**Before** executing the attack:

- Buy DNS domain
- Set up DNS server
- Develop software with C&C protocol

# Before Initial Access

- **Resource Development** Establish resources for supporting future operations
- Create, purchase, steal resources (software, infrastructure, accounts, capabilities) (7 techniques)

- Initial Access
- Execution
- Persistence
- C&C
- Discovery
- Lateral movement
- Exfiltration

# Even before…

- **Reconnaissance** Gather information for planning future operations (10 techniques)

- **Resource Development** Establish resources for supporting future operations
- Create, purchase, steal resources (software, infrastructure, accounts, capabilities) (7 techniques)

- Initial Access
- Execution
- Persistence
- C&C
- Discovery
- Lateral movement
- Exfiltration

# Defense:
# A Few Key Remarks

# Defense: A Few Key Remarks (I)

- Insisting on **complete prevention of Initial Access** is usually **meaningless** (perimeter just too large)
- Attacks **never** consist of **one** single step

- Defensive budget should be distributed across **all** attack phases
- A strong defense on a **few techniques** may suffice to **disrupt the attack** ("kill chain")

# Defense: A Few Key Remarks (II)

- Defensive budget should be distributed across **all** attack phases
- Defense must consist of:
  - **Mitigation**
    - "Prevent a technique from being successfully executed" = make attacks more difficult
  - **Detection**
  - **Remediation**
    - Backups

# Defense:
# A Few Key Remarks (III)

Techniques: 193

Sub-techniques: 401

- **Our job is very difficult**
  - **Real** complexity (not an ATT&CK artifact)
  - It is unlikely that we really understand all the techniques

- We need **systematic methods** for:
  - **Understanding** the **scope** of defensive mechanisms
  - **Prioritizing** techniques
  - Understanding the (potential) scope of **data sources**

# Understanding MITRE ATT&CK

# Attack vs MITRE ATT&CK ?

- **14** phases (called "**Tactics**")
- Several ways for executing each phase ("**Techniques**")

- Given a specific attack
- How is it mapped on Tactics and Techniques?

# Attack vs MITRE ATT&CK (I)

- **14** phases (called "**Tactics**")
- Several ways for executing each phase ("**Techniques**")

- **NO**:
  - Touching **all** the Tactics

- **YES**:
  - One or more Tactics may be **absent** (or **not observed**)

# Attack vs MITRE ATT&CK (II)

- **14** phases (called "**Tactics**")
- Several ways for executing each phase ("**Techniques**")

- **NO**:
  - Each Technique is used for a specific Tactic

- **YES**:
  - A Technique may be used for multiple Tactics

# Example

| Initial Access<br>10 techniques | Execution<br>14 techniques | Persistence<br>20 techniques | Privilege Escalation<br>14 techniques | Defense Evasion<br>43 techniques |
|---|---|---|---|---|
| Valid Accounts (0/4) | Windows Management Instrumentation | Valid Accounts (0/4) | Valid Accounts (0/4) | XSL Script Processing |
| Trusted Relationship | User Execution (0/3) | Traffic Signaling (0/2) | Scheduled Task/Job (0/5) | Weaken Encryption (0/2) |
| Supply Chain Compromise (0/3) | System Services (0/2) | Server Software Component (0/5) | Process Injection (0/12) | Virtualization/Sandbox Evasion (0/3) |
| Replication Through Removable Media | Software Deployment Tools | Scheduled Task/Job (0/5) | Hijack Execution Flow (0/12) | Valid Accounts (0/4) |
| Phishing (0/4) | Shared Modules | Pre-OS Boot (0/5) | Exploitation for Privilege Escalation | Use Alternate Authentication Material (0/4) |
| Hardware Additions | | Power Settings | | Unused/Unsupported Cloud |

## Valid Accounts

Sub-techniques (4)

Adversaries may obtain and abuse credentials of existing accounts

# Attack vs MITRE ATT&CK (III)

- **14** phases (called "**Tactics**")
- Several ways for executing each phase ("**Techniques**")

- **NO**:
  - **Single** flow of Tactics, left to right

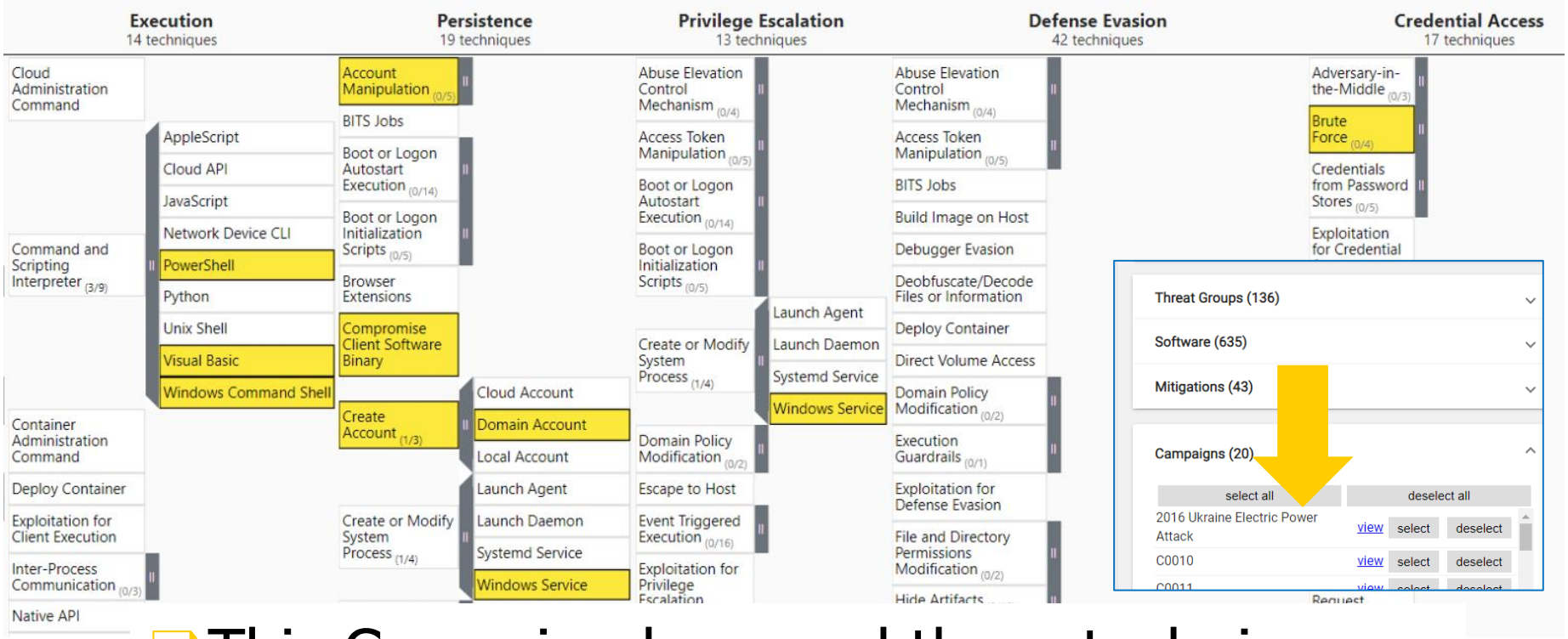
- **YES**:
  - **Multiple** flows/loops of Tactics, back and forth

# Example (I)

- …
- Discovery
- Lateral movement
- …
  - Machine M1 entered and controlled
  - Executing Discovery **again** usually provides further information…which may enable discovering M2
  - Machine M2 entered and controlled
  - Executing Discovery **again** usually provides further information…which may enable discovering M3

  - And in M2 / M3 you might need to execute Persistence again

# Example (II)




- This Campaign has used these techniques
- Order **not** apparent from the mapping

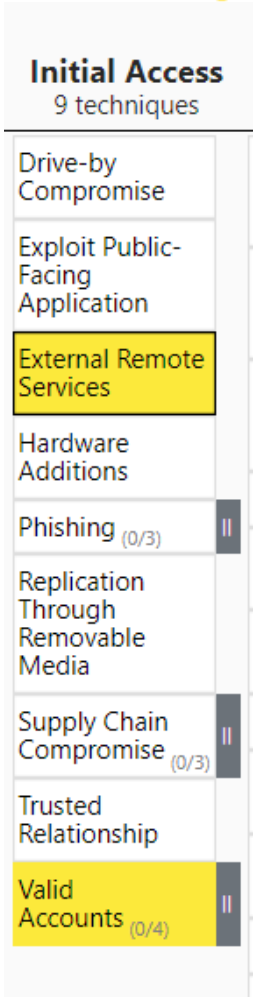
# Attack vs MITRE ATT&CK (IV)

- **14** phases (called "**Tactics**")
- Several ways for executing each phase ("**Techniques**")

- **NO**:
  - Every attack step clearly corresponds to **one** specific Technique

- **YES**:
  - Every attack step may correspond to one or **more** Techniques (even in **different Tactics)**

# Example




- Campaign that used multiple techniques for Initial Access

# What MITRE ATT&CK is (and is NOT)

# What MITRE ATT&CK is NOT (I)

- For any given **technique**, we do **not** have any clue about:
  - **Frequency** / **Probability** of usage

- There are statistics
- But in cybersecurity we **never** know their **coverage**
  - How many incidents missing from the statistics?

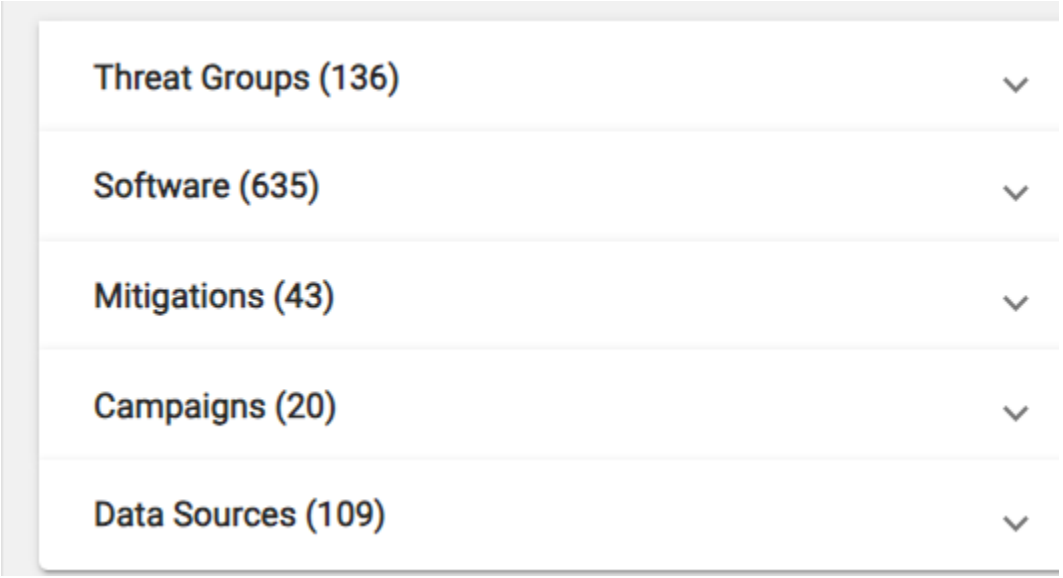
- ...nor their **bias**
  - Is the sample really relevant for "our" environment?

# What MITRE ATT&CK is NOT (II)

- For any given **technique**, we do **not** have any clue about:
  - **Frequency** / **Probability** of usage
  - Whether it is **absolutely essential** for a given attacker
    - Stopping this technique stops the attack?

# What MITRE ATT&CK is

- **Database** (with "links and navigation") for associating **tactics** / **techniques** with:



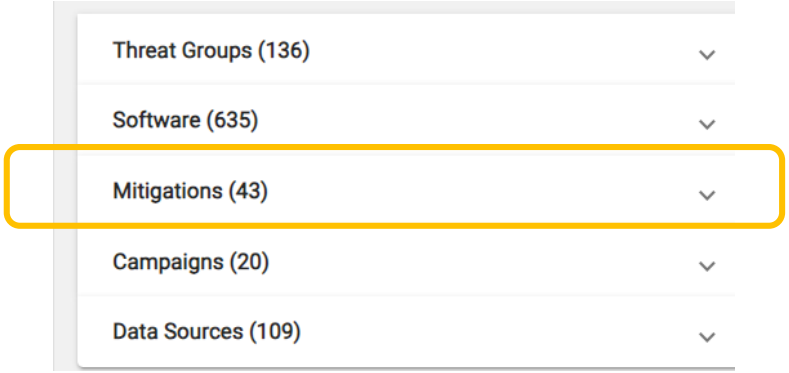

- Coverage obviously incomplete

# Example: Mitigations

Mitigations represent security concepts and classes of technologies that can be used to prevent a technique or sub-technique from being successfully executed.

| ID | Name | Description |
|---|---|---|
| M1036 | Account Use Policies | Configure features related to account use like login attempt lockouts, specific login times, etc. |
| M1015 | Active Directory Configuration | Configure Active Directory to prevent use of certain techniques; use SID Filtering, etc. |
| M1049 | Antivirus/Antimalware | Use signatures or heuristics to detect malicious software. |

- Which **techniques** are covered by a certain **mitigation**?
- Which **mitigations** exist for a certain **technique**?

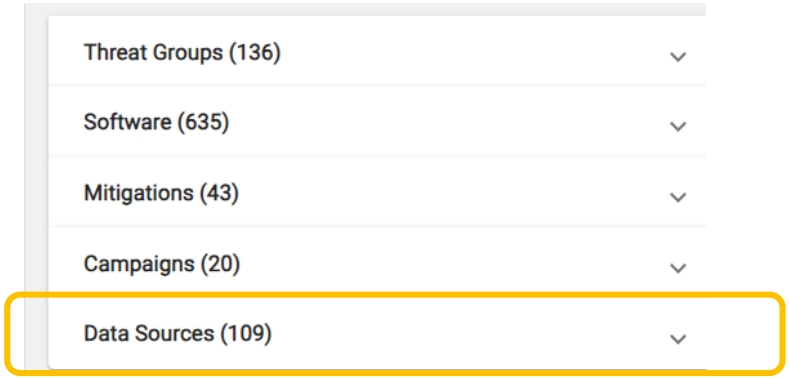


# Example: Data Sources (≈"log")

Data sources represent the various subjects/topics of information that can be collected by sensors/logs. Data sources also include data components, which identify specific properties/values of a data source relevant to detecting a given ATT&CK technique or sub-technique.

| ID | Name | Domain | Description |
|---|---|---|---|
| DS0026 | Active Directory | Enterprise | A database and set of services that allows administrators to manage permissions, access to network resources, and stored data objects (user, group, application, or devices) |
| DS0015 | Application Log | Enterprise ICS | Events collected by third-party services such as mail servers, web applications, or other appliances (not by the native OS or platform) |

- Which **techniques** could be detected by a certain **data source**?
- Which **data source** could enable detecting a certain **technique**?

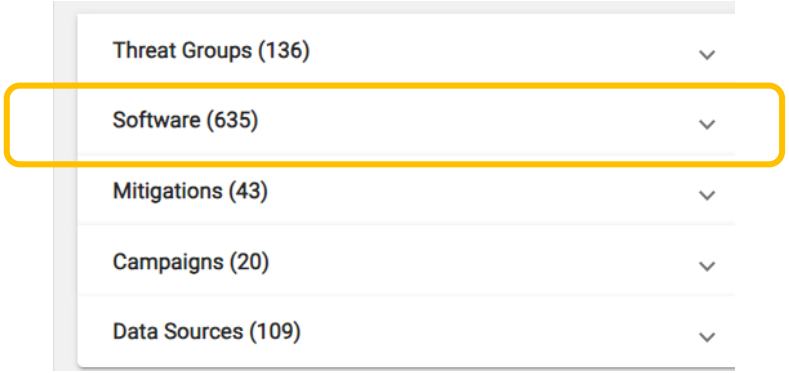


# Example: Software (I)

## CrackMapExec

CrackMapExec, or CME, is a post-exploitation tool developed in Python and designed for penetration testing against networks. CrackMapExec collects Active Directory information to conduct lateral movement through targeted networks.[1]

- ≈20 **techniques**



# Example: Software (II)

- **Identify** all machines in an IP address range
  `cme smb IP-range`

**Discovery**

- **Attempt credentials** on all machines
  `cme smb IP-range -u username -p password`
  `(-H password-hash)`

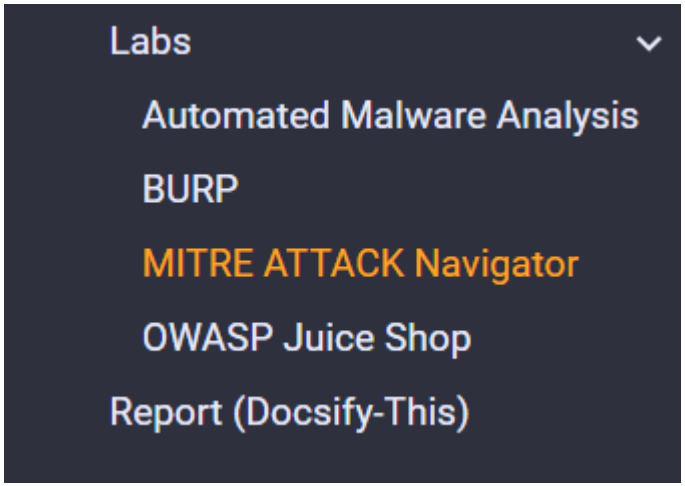
**Lateral Movement**

- **Extract password hashes** from all machines where local admin
  `cme smb IP-range -u username -p password`
  `-M mimikatz`

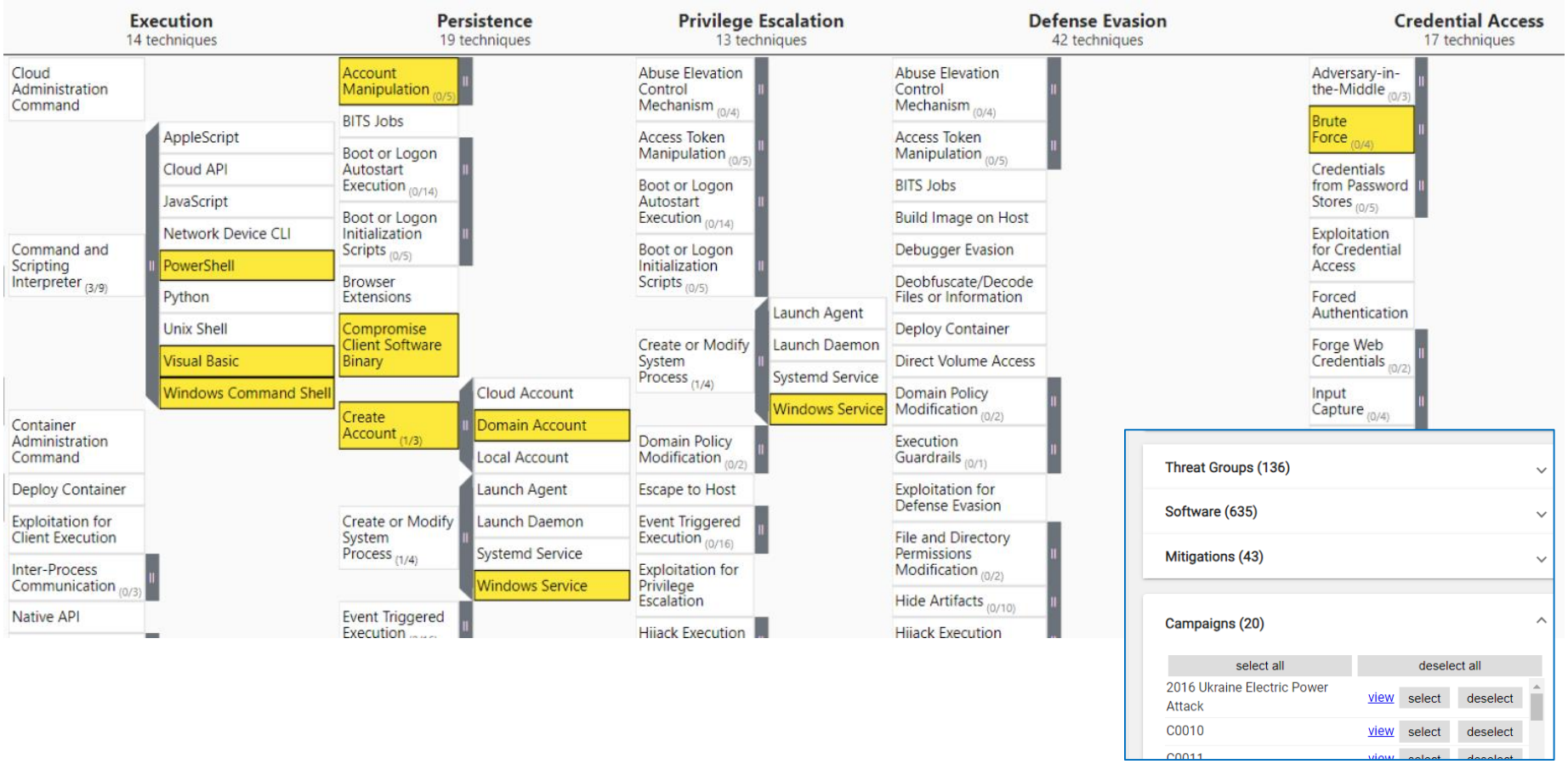
**Credential Access**

# Example: Navigator

- Which **techniques** are covered by **my mitigations**?
- Which **techniques** are used by a certain **threat group**?
- Which techniques am I **missing** w.r.t. to a certain threat group?



# Ukraine – Power Grid 2016 Campaign



# WARNING

- ATT&CK® Navigator (mitre-attack.github.io)
  (the software)
- Matrix - Enterprise | MITRE ATT&CK®
  (the official database)

- **Not** aligned perfectly

# Common Usage

- **Framework** for:
  - **Describing** attack campaigns
  - **Reasoning** about attacks and attack campaigns

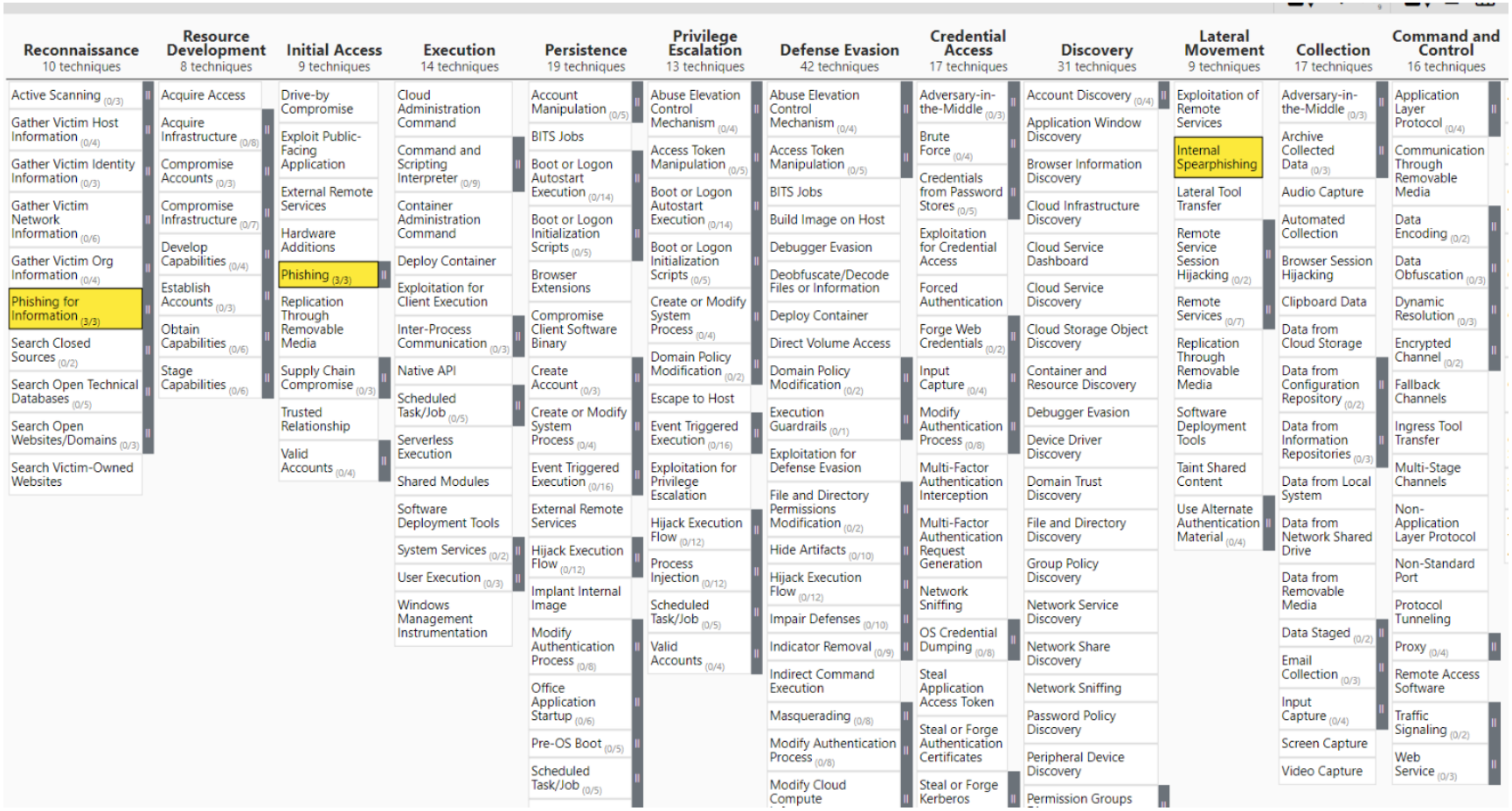
- **Very powerful (conceptual) tool**

# My suggestions

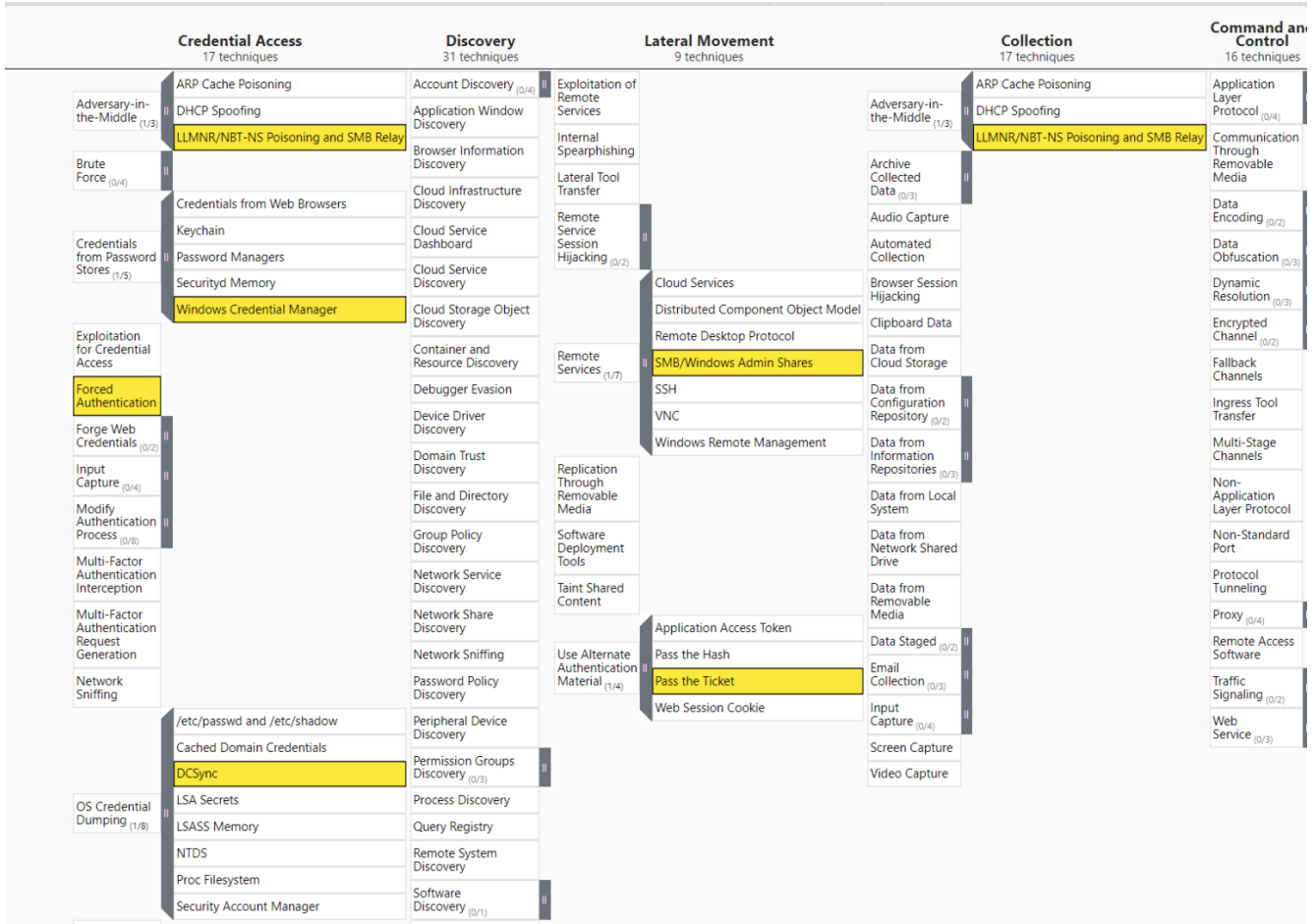
- For each topic covered in the course, **always try** to understand which **Tactic** (≈phase) it relates to

- Keep in mind that such a mapping may be **complex** and **not intuitive**
  - One topic may relate to multiple **Techniques** in different **Tactics**

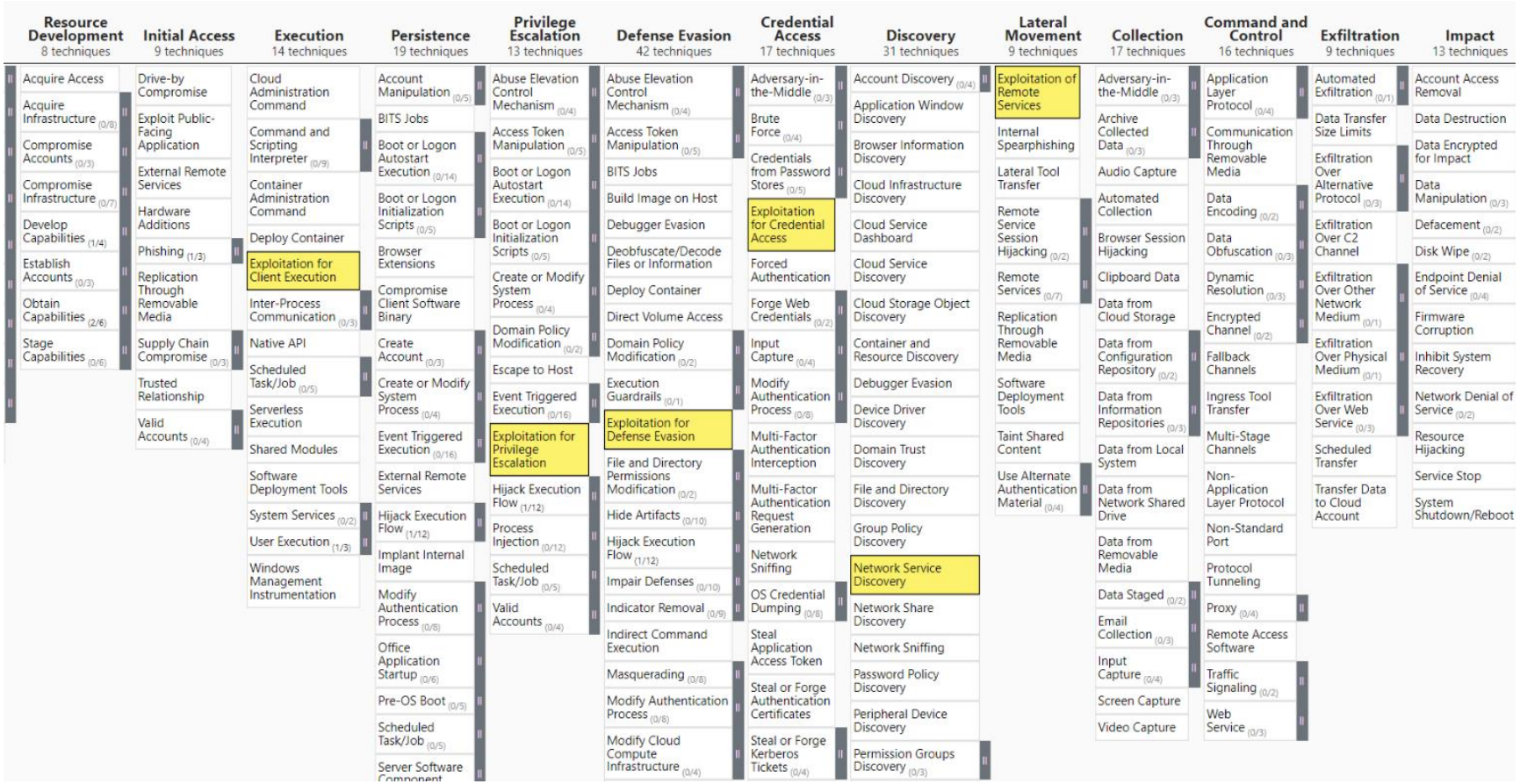
# Example: "phishing"



# Example: "ntlm"



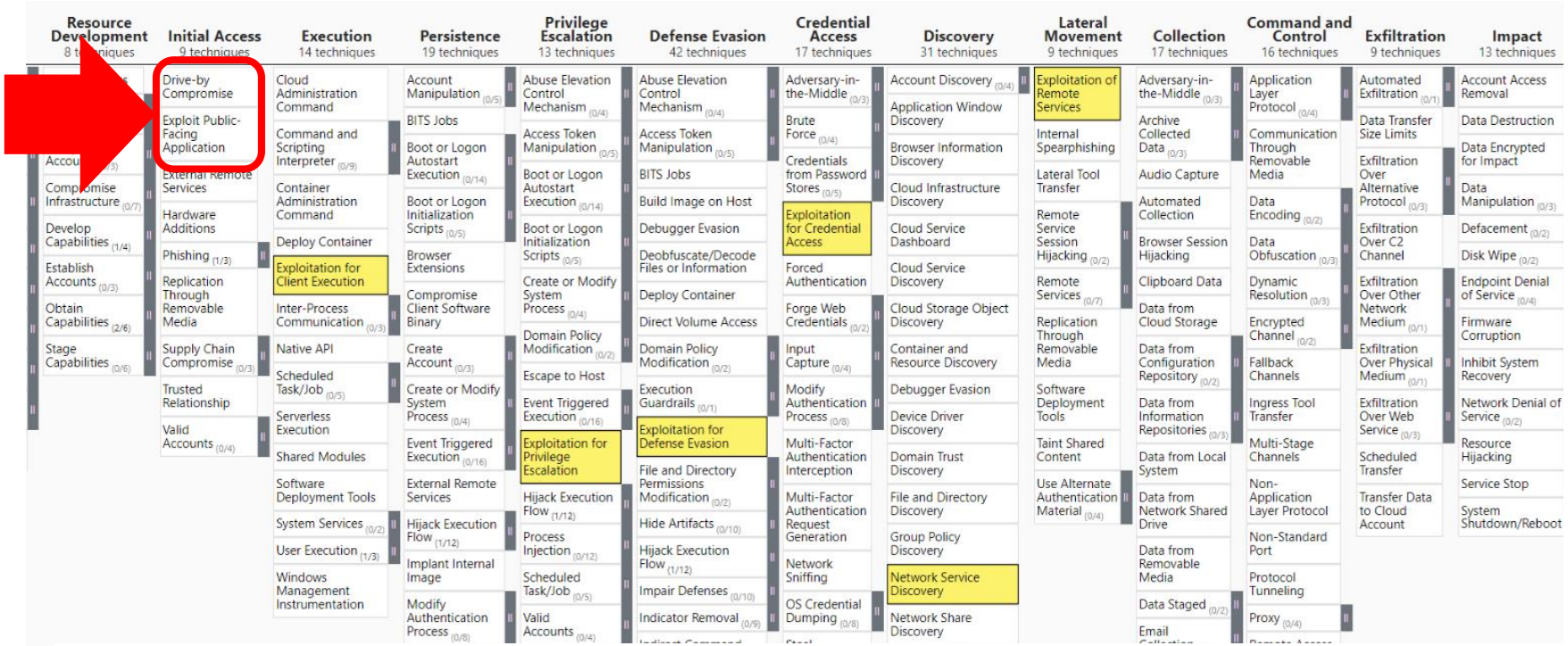
# Example: "vulnerability"



# Warning

- Keep in mind that such a mapping may be **complex** and **not intuitive**
  - One topic may relate to multiple **Techniques** in different **Tactics**
  - Relation topic-might **not** be encoded in ATT&CK
  - ...or it may follow criteria different from ours
    - Personal assessment often necessary

# Example: Vulnerability




Why not highlighted?

# Attacking an ICS

# Target Categories: ICS

1. Organizations
2. Industrial Control Systems (ICS)
3. Single individuals

- Administration
- Logistics
- Payroll
- Sales / Purchasing
- Warehouse
- ...
- Email / Web
- ...

Organization
("IT part of an industry")

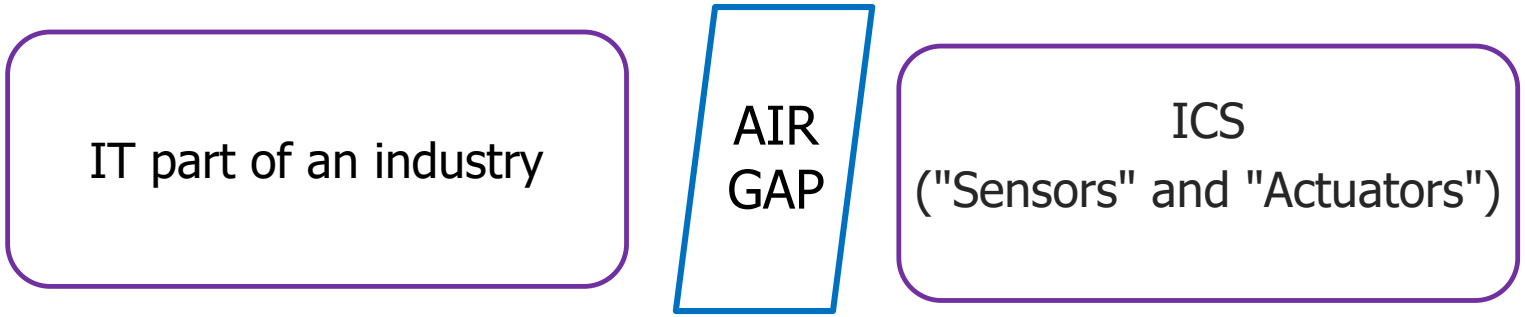
**OT**
**(Operational Technology)**

"Sensors" and "Actuators"

ICS

# Air Gap: Theory

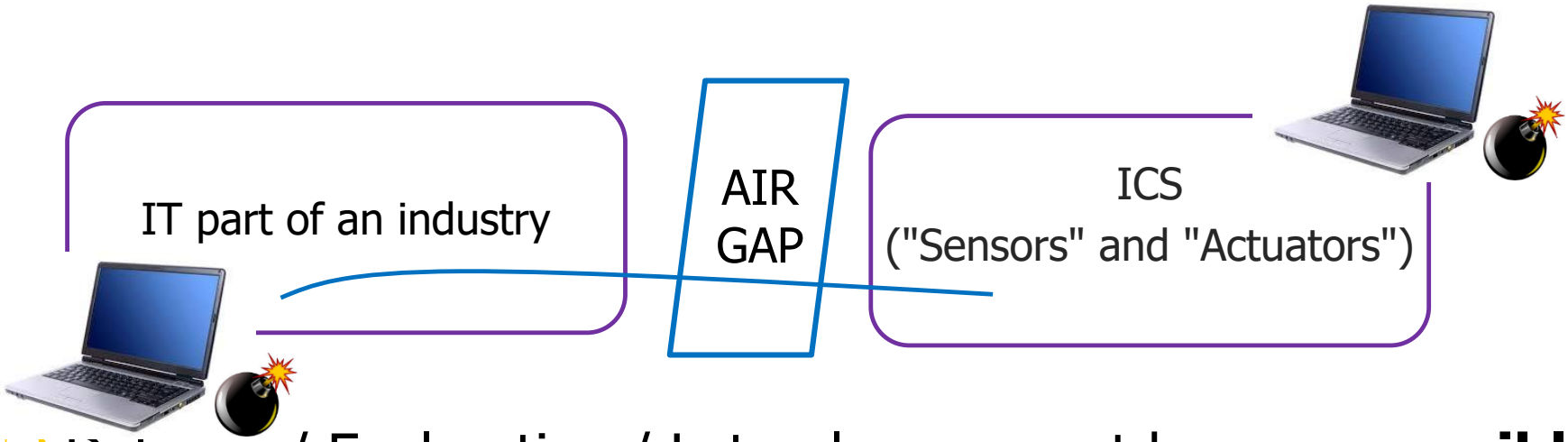
- IT part connected to the Internet
- ICS part **fully disconnected** from the IT part and from the Internet




- Delivery / Exploration / Lateral movement **not** possible

# Air Gap: Practice

- Support engineers occasionaly connect their notebooks on the ICS
- ICS permanently accessible from (selected locations of) IT part for remote control / monitoring
  - ...sometimes even from the Internet




- Delivery / Exploration / Lateral movement become **possible**

# MITRE ATT&CK Matrix

| Initial Access | Execution | Persistence | Privilege Escalation | Evasion | Discovery | Lateral Movement | Collection | Command and Control | Inhibit Response Function | Impair Process Control | Impact |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 12 techniques | 9 techniques | 6 techniques | 2 techniques | 6 techniques | 5 techniques | 7 techniques | 11 techniques | 3 techniques | 14 techniques | 5 techniques | 12 techniques |
| Drive-by Compromise | Change Operating Mode | Hardcoded Credentials | Exploitation for Privilege Escalation | Change Operating Mode | Network Connection Enumeration | Default Credentials | Adversary-in-the-Middle | Commonly Used Port | Activate Firmware Update Mode | Brute Force I/O | Damage to Property |
| Exploit Public-Facing Application | Command-Line Interface | Modify Program | Hooking | Exploitation for Evasion | Network Sniffing | Exploitation of Remote Services | Automated Collection | Connection Proxy | Alarm Suppression | Modify Parameter | Denial of Control |
| Exploitation of Remote Services | Execution through API | Module Firmware | | Indicator Removal on Host | Remote System Discovery | Hardcoded Credentials | Data from Information Repositories | Standard Application Layer Protocol | Block Command Message | Module Firmware | Denial of View |
| External Remote Services | Graphical User Interface | Project File Infection | | Masquerading | Remote System Information Discovery | Lateral Tool Transfer | Data from Local System | | Block Reporting Message | Spoof Reporting Message | Loss of Availability |
| Internet Accessible Device | Hooking | System Firmware | | Rootkit | Wireless Sniffing | Program Download | Detect Operating Mode | | Block Serial COM | Unauthorized Command Message | Loss of Control |
| Remote Services | Modify Controller Tasking | Valid Accounts | | Spoof Reporting Message | | Remote Services | I/O Image | | Change Credential | | Loss of Productivity and Revenue |
| Replication Through Removable Media | Native API | | | | | Valid Accounts | Monitor Process State | | Data Destruction | | Loss of Protection |
| Rogue Master | Scripting | | | | | | Point & Tag Identification | | Denial of Service | | Loss of Safety |
| Spearphishing Attachment | User Execution | | | | | | Program Upload | | Device Restart/Shutdown | | Loss of View |
| Supply Chain Compromise | | | | | | | Screen Capture | | Manipulate I/O Image | | Manipulation of Control |
| Transient Cyber Asset | | | | | | | Wireless Sniffing | | Modify Alarm Settings | | Manipulation of View |
| Wireless Compromise | | | | | | | | | Rootkit | | Theft of Operational Information |
| | | | | | | | | | Service Stop | | |
| | | | | | | | | | System Firmware | | |

- In a nutshell:
  - "General" tactics more or less the same
  - Two more tactics: Inhibit Response, Impair Process Control
  - Much less techniques

# Target Category: Organization

- Many **similarities** between Organizations
- A **given set** of skills, tools and knowledge is highly effective on **many different** organizations
- Standard, highly effective procedures for obtaining **money**

# Target Category: ICS

- **Very few similarities** between ICSs
- A **given set** of skills, tools and knowledge is highly effective on **very few** ICSs
- You need to **invent** some **highly specific** way for obtaining **money**

- Attacks to ICS are **much less frequent** than attacks to Organizations:
  - Much more costly
  - Much more difficult to get money

# Important Remark 1

- Attacks to ICS are much less frequent than attacks to organizations
  - Much more costly
  - Much more difficult to get money

- Attacks on ICS may have strategic / intelligence motivations
  - High budget
  - Objective is Data stealing / Disruption (not Money)

# Example 1

KIM ZETTER SECURITY MAR 3, 2016 7:00 AM WIRED

## Inside the Cunning, Unprecedented Hack of Ukraine's Power Grid

The hack on Ukraine's power grid was a first-of-its-kind attack that sets an ominous precedent for the security of power grids everywhere.

- ...about 30 substations offline...two other power distribution centers at the same time...leaving more than 230,000 residents in the dark.
- They also disabled backup power supplies...leaving operators themselves stumbling in the dark.

- Spear phishing then **many months** of extensive **reconnaissance**...
- Each company used a different distribution management system for its grid, and during the reconnaissance phase, the attackers studied each of them carefully.

# Example 2

Die Lage der IT-Sicherheit in Deutschland 2014

Bundesamt für Sicherheit in der Informationstechnik

- Targeted attack on a **steel mill** in Germany (pg. 31)
- There were frequent failures of individual control components or entire systems.
- …a **blast furnace was not regulated**, it could be shut down and get in an undefined state…
- As a consequence there was **massive damage** to the facility.

# Example 3

## Alert (AA22-083A)

### Tactics, Techniques, and Procedures of Indicted State-Sponsored Russian Cyber Actors Targeting the Energy Sector

Original release date: March 24, 2022

CYBERSECURITY & INFRASTRUCTURE SECURITY AGENCY

- Multiple intrusion campaigns conducted by **state-sponsored Russian cyber actors from 2011 to 2018** and targeted U.S. and international **Energy** Sector

- Description with MITRE ATT&CK framework
  `https://bartoli-alberto.blogspot.com/search?q=guerra`

# Important Remark 2

- Attacks to ICS are much less frequent than attacks to organizations
  - Much more costly
  - Much more difficult to get money

- An attack on the **"IT part"** may **disrupt** industrial operations

# Example 1

*Cyberattack Forces a Shutdown of a Top U.S. Pipeline*

The New York Times

May 13, 2021

The operator, Colonial Pipeline, said it had halted systems for its 5,500 miles of pipeline after being hit by a ransomware attack.

- One of the nation's largest pipelines, which carries refined gasoline and jet fuel from Texas up the East Coast to New York, was forced to shut down after being hit by ransomware...
- Colonial Pipeline...had shut down its 5,500 miles of pipeline, which it says carries 45 percent of the East Coast's fuel supplies, in an effort to contain the breach.

# Example 2

## Toyota halts operations at all Japan plants due to cyberattack

NIKKEI Asia

February 28, 2022

- Toyota Motor on Tuesday halted operations at all of its plants in Japan after a major supplier was hit by a cyberattack, disrupting the automaker's parts supply management system.

# Example 3

## NonPetya ransomware forced Maersk to reinstall 4000 servers, 45000 PCs

ZDNet

The shipping giant has suffered millions of dollars in damage due to the ransomware attack. January 26, 2018

- Maersk has revealed that a devastating ransomware attack which struck businesses across Europe in 2017 required close to a "complete infrastructure" overhaul and the reinstallation of thousands of machines.
- The firm, with offices in 130 countries and a workforce of close to 90,000,

- "Imagine a company where a ship with 10 to 20 thousand containers is entering a port every 15 minutes, and for 10 days, you have no IT," Hagemann commented. "It's almost impossible to even imagine."

# Key remarks

- Computer attacks no longer affect only "**data**"
- They may affect the "**physical world**"
- They may **disrupt** "**non IT** orgs"

# Attacking Single Individuals

# Target Categories: Single Individuals

1. Organizations
2. Industrial systems (ICS)
3. Single individuals

- Initial Access
- Execution
- Persistence
- C&C
- ~~Discovery~~
- ~~Lateral movement~~
- Impact

# Motivations

1. Money
2. Stealing of information
3. Disruption of operations

- Money is by far the **most frequent** motivation

- Look at "How to obtain money"

# Key Remark

- Human operators execute **all the steps**
- Actions can be **tailored** to the **specific** environment
- Costly

- Automated tool executes **all the steps**
- Actions **cannot** be **tailored** to the **specific** environment
- Investment can be amortized over many targets

- **Automation is much more frequent**
  - Can be made **very effective**
    (unlike attacks to organizations)
  - Only way for justifying **small gain** per successful target
    (attacks to organization have large gain per successful target)